Session 3: Git and LaTeX in VS Code

Version Control and the Integrated Research Environment

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Where We Are

Session 1: What Claude Code can do. How it works. CLAUDE.md and plan mode.

Session 2: The core workflow. Data cleaning, estimation, robustness, LaTeX tables.

Session 3: Building infrastructure around the workflow.

Today you will:

1. Learn the minimum viable git workflow for agentic coding
2. Bring your paper into the same project directory as your code
3. Use Claude Code to write, compile, and version-control a LaTeX document

Session 2 taught you to verify Claude Code's work. Today you get the tools to manage it.

Git is research insurance,
not software engineering ceremony.

You need exactly four commands.

What Are Git and GitHub?

Git is a tool that tracks every change to every file in a project. It runs on your computer. It is free and open source.

Think of it as "track changes" for your entire project directory — not just one document, but every file: code, data descriptions, LaTeX files, everything.

GitHub is a website that hosts git projects online.

- It is where open-source code, replication packages, and increasingly papers live
- Git works entirely on your machine. GitHub is optional — it adds backup, sharing, and collaboration
- Today we only use git locally. GitHub comes in Session 4.

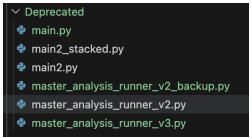
The Key Idea

Git lets you save snapshots of your project at any point. You can always see what changed between snapshots, and go back to any

Why Researchers Should Care About Git

Without git, you have probably done this:

- `analysis_v2.py`, `analysis_v2_final.py`, `analysis_v2_final_REAL.py`
- "Which version of the cleaning script produced these results?"
- "I changed something last week and now the code is broken"



With git:

- One file, full history. Every version is recoverable.
- You can see exactly what changed, when, and why.
- You can undo any change — even from weeks ago.

Git is useful for any research project. But when you are working with an agent that modifies your files directly, it goes from useful to essential.

Why Git Matters for Agentic Coding

The problem: Claude Code modifies your files directly.

- It might change a file you did not ask it to change
- It might overwrite working code with something broken
- It might make 10 changes when you only wanted 3

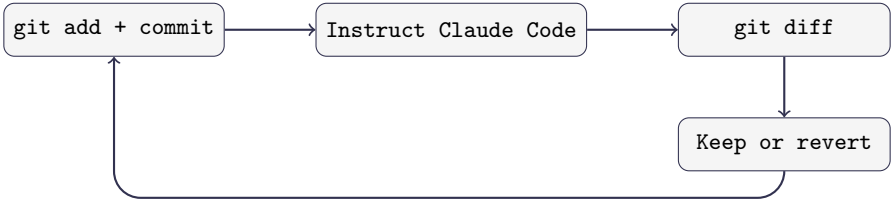
Without git: You try to manually undo changes. You are not sure what the file looked like before. You lose work.

With git: You see exactly what changed, line by line. You revert anything you do not want. You never lose work.

The Rule

Never let Claude Code touch your project without a recent commit to fall back on.

The Minimum Viable Git Workflow



Command	What it does
`git add -A && git commit -m "msg"`	Snapshot your current state
`git diff`	See what changed since last commit
`git checkout -- file.py`	Undo changes to a specific file
`git log --oneline`	See your commit history

Command 1: `git init`

What it does: Turns a regular folder into a git project.

```
git init
```

- You run this **once**, when you start a new project
- It creates a hidden `.git/` folder that stores your project's history
- Nothing else changes — your files stay exactly as they are

Think of it as saying: "I want to start tracking this folder."

Note

If you cloned this course repository, git is already initialized. You can check by running `git status` — if it prints something, you are already set up.

Command 2: `git add`

What it does: Selects which files to include in your next save point.

```
git add -A
```

- `-A` means "add everything that changed" — new files, modified files, deleted files
- This does not save anything yet. It just marks what will be included in the next commit.

You can also add specific files instead of everything:

```
git add scripts/analysis.py
```

Think of it as putting files in a box before sealing it. `git add` fills the box; `git commit` seals it.

Command 3: `git commit`

What it does: Saves a snapshot of everything you added.

```
git commit -m "Clean merge script, verified row counts"
```

- `-m` stands for "message" — a short note describing what this snapshot contains
- The message goes in quotes. Write something your future self will understand.
- After committing, you have a permanent save point you can always return to.

In practice, you can always run add and commit together (but you don't have to):

```
git add -A && git commit -m "your message"
```

The `&&` just means "run the second command after the first one finishes."

Command 4: `git diff`

What it does: Shows you exactly what changed since your last commit.

```
git diff
```

The output looks like the diff view you already know from VS Code:

- Lines starting with + were added (green in VS Code)
- Lines starting with − were removed (red in VS Code)
- Everything else is unchanged context

When to use it: After Claude Code makes changes, run `git diff` to see *everything* it touched — not just the file it told you about.

Tip

You can also ask Claude Code: "Run git diff and explain each change you made."

Command 5: `git checkout`

What it does: Undoes changes by restoring a file to its last committed state.

Undo changes to one file:

```
git checkout -- scripts/analysis.py
```

This throws away all changes to that file since your last commit. The file goes back to exactly how it was.

Undo all changes (reset everything):

```
git checkout -- .
```

The `.` means "this directory" — it reverts every file in your project.

This is your undo button

Claude Code changed something you did not want? One command and it is gone. This only works if you committed first — which is why you always commit before instructing.

Command 6: `git log`

What it does: Shows you the history of all your commits.

```
git log --oneline
```

Output looks like:

```
a3f7b21 Add robustness checks
e5c9d04 Clean merge script, verified row counts
b1a2c33 Initial commit
```

- Each line is one commit — one save point
- The letters and numbers on the left are the commit ID
- The message on the right is what you wrote with `-m`
- Most recent commits are at the top

This is your project's timeline. You can always see what you did and when.

Putting It Together: The Cycle

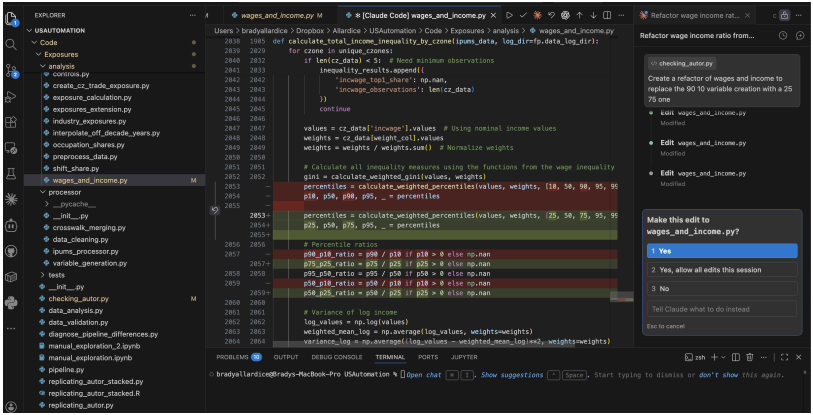
Before every interaction with Claude Code:

1. `git add -A && git commit -m "message"` — save your current state
2. Give Claude Code an instruction
3. `git diff` — see what it changed
4. Decide:
   - Happy? → `git add -A && git commit -m "message"`
   - Not happy? → `git checkout -- .`
5. Repeat

The Mindset Shift

Commits are not milestones. They are save points — like pressing Ctrl+S. Commit early, commit often, commit before every interaction with Claude Code.

You Have Already Seen This



Red = removed, green = added. You have been reading diffs since Session 1.

In VS Code: When working directly in the editor, you see the `git diff` visually before you accept the change. This shows exactly what Claude Code changed.

Git and Data: What Not to Track

Git is designed for **code and text files** — things that change line by line.
It is *not* designed for large data files.

Commit:

- Scripts (`.py`, `.R`, `.do`)
- LaTeX files (`.tex`, `.bib`)
- Documentation (`CLAUDE.md`)
- Small data files (under a few MB)

Do not commit:

- Large datasets (50MB+)
- Output you can regenerate
- Sensitive data (PII, API keys)

Rule of Thumb

Commit what a collaborator needs to *understand* your project. Skip what they could *regenerate* by running the code.

.gitignore: Telling Git What to Skip

A .gitignore file lists patterns for files git should never track. Place it in the folder where you ran git init.

Paths are relative to that folder. Example:

```
my_project/ <- run "git init" here
  .gitignore <- goes here
  code/
    analysis.py
  data/
    raw/ <- "data/raw/" in .gitignore
    processed/
  output/ <- "output/" in .gitignore
```

Always include:

```
.DS_Store
.env
*.pyc
__pycache__/
*.aux
```

So Where Does the Data Go?

Why git is bad for data:

- Git stores *every version* of every file. A 200MB CSV edited 10 times = 2GB of history.
- Git compares files line by line. Change one cell in a CSV and git sees the entire row as new.
- Large files slow down every git operation — commits, diffs, cloning.

For most research projects (data under a few GB):

- Keep data in a synced folder: **Dropbox**, **OneDrive**, **Google Drive**
- Your code lives in git. Your data lives next to it. Your `.gitignore` keeps them separate.
- Simple, familiar, and good enough for most social science workflows

For larger or more complex needs:

- **Object storage:** AWS S3, Google Cloud Storage — cheap, scalable
- **Data versioning:** DVC (Data Version Control) — git-like tracking for large files

You do not need these today. But if your data outgrows Dropbox, they exist.

Exercise Part 1: Initialize and Commit (10 min)

Choose one:

- **Option A:** Work in your own research project
- **Option B:** Use the provided Session 2 materials as a practice project (if you don't want to touch your own work)

1. **Create a** `.gitignore` for your project (Ask Claude Code to help):
   - Large data files: `*.csv.gz`, `*.parquet`, `data/raw/`
   - Output that regenerates: `output/`, `*.pdf`
   - System files: `.DS_Store`, `.env`, `*.pyc`
2. **Initialize git**:
   - `git init`
   - `git add -A && git commit -m "Initial commit with .gitignore"`
3. **Run** `git log --oneline` to see your history.

You now have a snapshot of everything that works. Whatever happens next, you can always come back here.

Exercise Part 2: The Cycle (25 min)

Do this **at least three times:**

1. **Commit** your current state
2. **Instruct Claude Code** to make a change:
   - Round 1: Add a new control variable to your specification
   - Round 2: Refactor the analysis script into functions
   - Round 3: Change the sample restriction
3. **Run** `git diff` and read what changed
4. **Decide:** keep (commit) or revert (`git checkout -- .`)

Try to revert at least once. The point is to see that you can always undo.

Commit first, instruct second, diff third, decide fourth.

If you take one thing from this module, it is this sequence.

Git Debrief

Discussion:

- Did Claude Code change any files you did not expect?
- Was the diff easy to read? What was confusing?
- Did anyone revert a change? What went wrong?

The Habit

Commit → Instruct → Diff → Decide.

This is not optional when working with an agent that modifies your files. It is the minimum responsible workflow.

In Session 4, we will automate the commit step with a hook — so you never forget.

What if your paper lived in the same place as your code and data?

This is the conceptual pivot of the course.

The Problem with Overleaf

Most researchers keep their paper in a separate tool.

- Code lives in a project directory (or on your Desktop)
- Data lives next to the code (maybe)
- The paper lives in Overleaf (or Word, or Google Docs)

What this means for Claude Code:

- It cannot read your paper
- It cannot check the methods section against your code
- It cannot update a table when the analysis changes
- It cannot commit changes to your paper alongside code changes

The Limitation

If Claude Code cannot see it, Claude Code cannot operate on it.
Your paper is the biggest piece of your project that is currently invisible.

The Solution: LaTeX in VS Code

Bring your paper into the project directory.

- Your `.tex` file lives alongside `scripts/` and `data/`
- Claude Code can read it, write it, and edit it — just like any other file
- Changes to the paper are tracked by git, just like code changes
- Coauthors review changes through version control, not "track changes"

What becomes possible:

- Claude Code writes a methods section and you verify it against the code
- The analysis changes $\rightarrow$ the table updates $\rightarrow$ the paper updates
- Every version of the paper is recoverable through git

The Point

This is not about LaTeX vs. Word. It is about bringing your paper into the agentic workflow.

LaTeX in VS Code: The Setup

What you need (should be installed from pre-work):

1. A TeX distribution: TeX Live (Linux/Windows) or MacTeX (Mac)
2. The **LaTeX Workshop** extension in VS Code

What you get:

- Syntax highlighting for `.tex` files
- One-click compilation to PDF
- PDF preview in a split pane — side by side with your code
- Error messages in the VS Code terminal

The workflow:
Edit `.tex` → Save → Auto-compile → PDF updates in the preview pane.

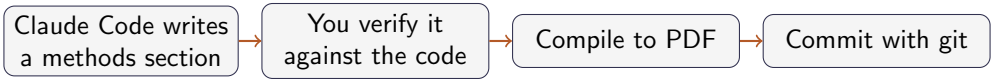
You never leave VS Code. Your paper is right next to your code.

LaTeX Basics (For Those New to It)

```
\documentclass{article}
\usepackage{booktabs}
\title{My Research Paper}
\author{Your Name}
\begin{document}
\maketitle
\section{Data Description}
We use a survey of 2,044 Polish adults...
\section{Results}
\input{output/robustness_table.tex}
\end{document}
```

Key idea: `\input{output/robustness_table.tex}` pulls in the table from Session 2. When the table updates, the paper updates.

Demo: The Full Loop



Everything happens in VS Code.

Everything is version-controlled.

Everything is visible to Claude Code.

Exercise Part 3: Build a LaTeX Document (30 min)

Using the provided template:

1. **Commit** your current state (the habit starts now)
2. **Ask Claude Code** to create a `.tex` file with:
   - A data description section based on your Session 2 pipeline
   - The robustness table via `\input{output/robustness_table.tex}`
3. **Compile** to PDF and verify it renders correctly
4. **Diff and commit** the new file
5. **Ask Claude Code** to write a results paragraph interpreting the table. Review what it writes — would you sign off on this for a paper?
6. **Ask Claude Code** to check the data description against the actual cleaning code. Does the text match what the code does?

Notice: you are now using git and LaTeX together. Every change to the paper is tracked.

What You Built Today

1. **Git:** You have a safety net. Commit → instruct → diff → decide. You practiced reverting changes and saw that you can always go back.
2. **LaTeX in VS Code:** Your paper is now part of your project. Claude Code can see it, edit it, and check it against your code.
3. **The two together:** Every change to your paper is version-controlled, just like every change to your code. You used the git cycle on your `.tex` file.

The Bigger Picture

Your project directory now contains: data, code, output, and a paper — all under version control, all visible to Claude Code. This is the foundation for everything in Session 4.

Debrief

Discussion questions:

1. Did Claude Code change any files you did not expect? How did you handle it?
2. Did anyone revert a change? What went wrong?
3. When Claude Code wrote parts of the LaTeX document, what did you have to correct?
4. How does this compare to your current workflow — Overleaf, Word, separate directories?

Key Takeaway

The skill is not just using Claude Code. It is building an environment where Claude Code can do its best work — and where you can verify and undo everything it does.

Next Session

Session 4: Power Features — Hooks, MCPs, and Skills

- **Hooks:** Automate the commit step — every change Claude Code makes is tracked automatically
- **MCPs:** Connect Zotero — search your library and pull citations from inside Claude Code
- **Skills:** Reusable workflows — spec validation, robustness checks, and more

Before next session:

- Practice the commit-instruct-diff-decide cycle on your own project
- Try asking Claude Code to review a piece of your own code
- Make sure your Zotero account is set up (instructions in the setup guide)